I. Preliminaries

1. Install R and RStudio. The following resources are helpful introductions to coding in R:

- R for Data Science (2e);
- A ModernDive into R and the Tidyverse;
- RYouWithMe;
- R Recipes.

2. For the subsequent exercises, you will need to install and load the following packages (in no particular order):

- data.table (for data-wrangling);[1]
- dplyr (for data-wrangling);
- tidyr (for data-wrangling);
- MASS (for density estimation);
- ggplot2 (for plotting);[2]
- ggdensity (for plotting);
- plotly (for plotting);
- magrittr packages (for better code readability).

II. Download Data

3. Set RStudio's working directory to the folder containing the datasets for this homework and open the PJM data set ("data_PJM.csv"), as below. This dataset is constructed from hourly pricing data from PJM's Data Miner 2 and includes both day-ahead ("DA") and real-time ("RT") locational marginal pricing ("LMP") data.

```
data_PJM <- fread('PJM Data.csv')
```

4. Load the following data set, which combines pricing and weather data for Chicago,[3] Milwaukee,[4] and Corpus Christi.[5]

```
data_Chicago <- fread('Data Chicago.csv')
data_Milwaukee <- fread('Data Milwaukee.csv')
data_CorpusChristi<- fread('Data Corpus Christi.csv')
data_Austin <- fread('Data Austin.csv')
```

[1] Code examples in this course will draw primarily on data.table tools, which can be better for handling larger datasets, but dplyr, from the Tidyverse, can be more intuitive for those new to R.
[2] *See also*, R Graphics Cookbook, 2nd edition.
[3] Pricing data from Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition. Weather data from Hitachi Energy Velocity Suite.
[4] Pricing data from MISO Market Reports; DA LMPs from Historical Annual Day-Ahead LMPs and RT LMPs from Historical Annual Real-Time LMPs. Weather data from Hitachi Energy Velocity Suite.
[5] Weather and LMP data from Hitachi Energy Velocity Suite.

III. Data Familiarization

5. Let's explore the PJM data! The following code identifies the fields of the dataset

```
> names(data_PJM)
 [1] "datetime_beginning_utc" "datetime_beginning_ept" "pnode_id"
"pnode_name"             "voltage"                "equipment"
 [7] "type"                   "zone"                   "system_energy_price_rt"
"total_lmp_rt"           "congestion_price_rt"    "marginal_loss_price_rt"
[13] "system_energy_price_da" "total_lmp_da"           "congestion_price_da"
"marginal_loss_price_da" "spread"                  "date"
[19] "hour"                   "am_pm"                  "day"
"day_of_year"            "month_num"              "month"
[25] "year"                   "day_of_week"            "weekend"
"isHoliday"              "holiday"
```

6. To see the first few observations of the data, run the code below. *What is the unit of observation?*

```
head(data_PJM)
```

7. The code below identifies the final date of the PJM data as December, 12, 2023. *How would you identify the initial date?*

```
> max(data_PJM$date)
[1] "2023-12-31"
```

8. The code below identifies the set of pricing node types in the PJM data and calculates how many observations correspond to each type. *How would you do the same for pricing nodes names?*

```
> unique(data_PJM$type)
[1] "ZONE"                 "AGGREGATE"            "HUB"                  "EHV"
"INTERFACE"            "RESIDUAL_METERED_EDC"
> data_PJM[,.N, by = type]
                    type       N
1:                  ZONE  578160
2:             AGGREGATE 4135890
3:                   HUB  315360
4:                   EHV 2795809
5:             INTERFACE  187583
6: RESIDUAL_METERED_EDC  709560
```

9. The code below identifies the number of unique holiday values. *How would you identify the number of observations falling on holidays? How would you identify the number of unique different pricing nodes?*

```
length(unique(data_PJM$holiday))
```

10. The following code restricts the PJM data to only the "date" and "total_lmp_rt" fields. *What fields would be interesting to study in conjunction?*

```
data_PJM[, .(date, total_lmp_rt)]
```

11. The code below creates a new dataset, restricted to PJM observations falling on Christmas. *Create a dataset, named "holidays_PJM," consisting of all observations falling on a holiday.*

```
xmas_PJM <- data_PJM[holiday == 'Christmas']
```

12. The code below calculates the average RT LMP on holidays in November and December, during AM hours, for the "'CHICAGO GEN HUB" pricing node.[6] *What observations are being selected? What is being averaged over? What stands out about the results? Write code that averages DA LMP within each month, for the "Chicago Gen Hub." Formulate a hypothesis based on the result you observe. How could you check your hypothesis?*

```
holidays_PJM[month == 'Dec' &
               pnode_name == 'CHICAGO GEN HUB' &
               am_pm == 'AM',
.(average_lmp_rt = mean(total_lmp_rt)), by = .(hour, year)] %>%
.[order(year)]
```

[6] Note that the pipe operator `%>%` from magrittr is helpful for chaining data-manipulation steps together and can be input using a shortcut (ALT + SHIFT + M).

IV. Introduction to Plotting

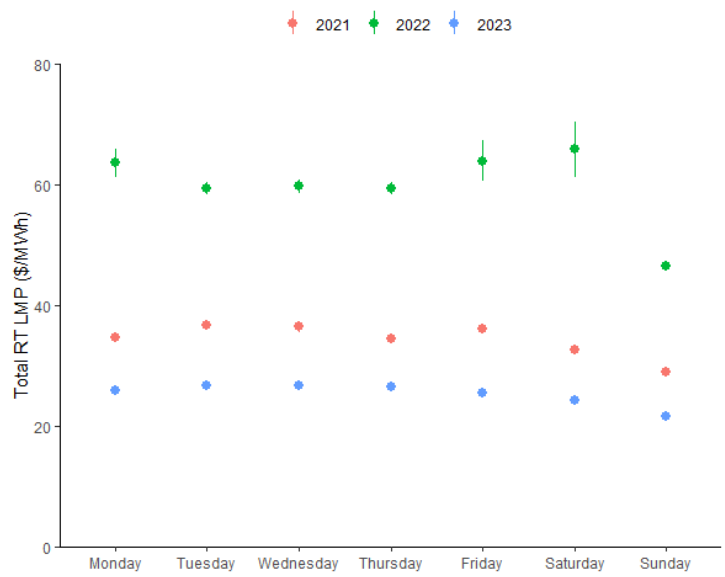
13. Create a dataset called "data_CGH," restricted to the "'CHICAGO GEN HUB." *What does the number of observations reflect?*

14. The code below generates **Figure 1**.[7] *What stands out? What is each point on the plot averaging over? Interpret the resulting figure when* `color = factor(year)` *is removed.*[8] *Try plotting RT LMP by month and by hour instead of day-of-week.*

```
data_CGH$day_of_week <- factor(data_CGH$day_of_week, levels =
c("Monday","Tuesday","Wednesday","Thursday","Friday","Saturday",
"Sunday"))

ggplot(data_CGH[order(day_of_week)], aes(x=day_of_week,y=total_lmp_rt, color =
factor(year))) +
  stat_summary() +
  labs(color = '', x = '', y = 'Total RT LMP ($/MWh)') +
  coord_cartesian(ylim = c(0, 80)) +
  scale_y_continuous(expand = c(0,0)) +
  theme_classic() +
  theme(legend.position = 'top')
```

Figure 1. Real-Time LMP by Day-of-Week (Chicago Gen Hub)[9]



15. Write code that plots average monthly day-ahead LMP for "CHICAGO GEN HUB." *What happened during the summer of 2022?*

16. Oftentimes, pivoting data from "wide" to "long" is helpful for plotting purposes. The follwing, for example, pivots the "data_CGH" by: [1] creating a "lmp_type" column that

[7] Note that the first block of code orders the days, so that the days of the weeks are in chronogical order (rather than alphabetical). For more on stat_summary, see here.
[8] Not all lines of the second block of code are necessary, although they render a more reader-friendly figure. Try deleting a few to better understand their functionality.
[9] Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.

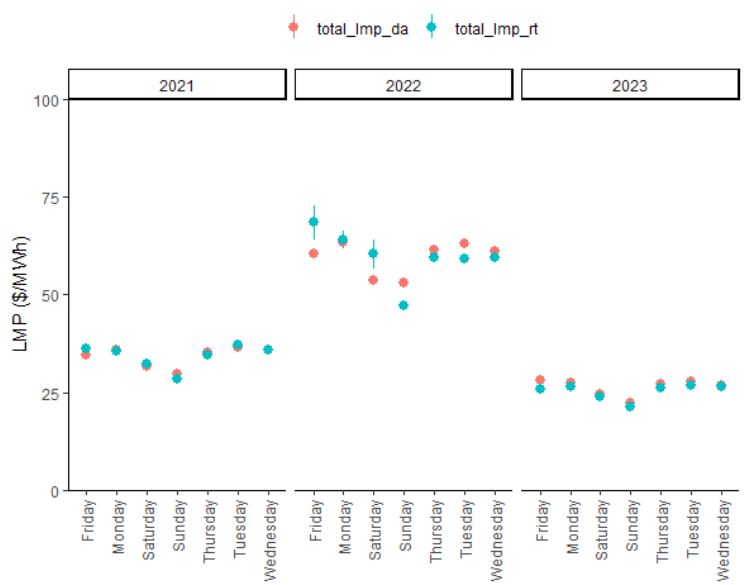
stores the names "total_lmp_rt," "total_lmp_da," and "prev_lmp_da"[10] as entries; and [2] creating a corresponding "values" column that stores values of those series.[11] *The second block of code plots and compares the "total_lmp_rt" series and "total_lmp_da" series by day-of-the-week, as well as by year. What happens when you replace "day_of_week" with month?*

```
data_CGH_long <- data_CGH[, .(year, month, hour, day_of_week, total_lmp_rt,
total_lmp_da)] %>%
  pivot_longer(cols = !c(year, month, hour, day_of_week),
               names_to = 'lmp_type',
               values_to = 'values')

setDT(data_CGH_long)

ggplot(data_CGH_long, aes(x=day_of_week,y=values, color = factor(lmp_type))) +
  stat_summary() +
  labs(color = '', x = '', y = 'LMP ($/MWh)') +
  coord_cartesian(ylim = c(0, 100)) +
  scale_y_continuous(expand = c(0,0)) +
  theme_classic() +
  theme(legend.position = 'top',
        axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1)) +
  facet_wrap(~year)
```

Figure 2. Day-Ahead vs. Real-Time LMP by Year and Day-of-Week (Chicago Gen Hub) [12]



17. The following code calculates average RT and DA LMPs for each day in the CGH dataset and then plots the empirical cumulative distribution (ECDF) of average RT LMPs by year.

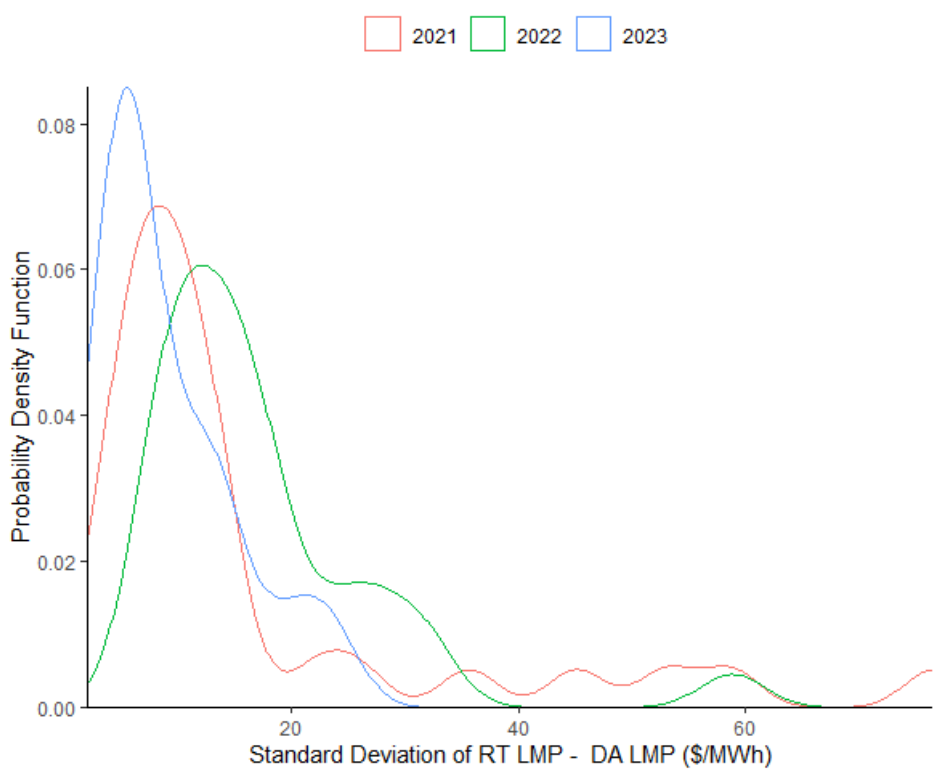
[10] "prev_lmp_da" is equal to "total_lmp_da," lagged by 24 hours.
[11] Put differently, a table with a single "year" column is the long version of a table with different years (e.g., "2021," "2022," and "2023") as columns. Note that the final line of code is necessary to reconvert our dataset into a data.table format, as the "pivot_longer" function creates a tidyr object, on which data.table syntax will no longer work.
[12] Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.

Note that the series corresponding to 2022 is below that for 2021 and 2023, indicating that for each possible value x, there were more days in 2021 and 2023 for which the average daily RT LMP was less than x; in other words, RT LMP was higher in 2022 than in 2021 or 2023. This relationship between probability distributions is known as first-order stochastic dominance (FOSD). *Adjust the code below such that it also calculates the daily volatility of RT LMP, using the sd() function, and identify another example of FOSD.*

```
CGH_summary_daily <- data_CGH[, .(mean_rt_lmp = mean(total_lmp_rt),
                                  mean_da_lmp = mean(total_lmp_da)),
                              by = .(day, month,year)]

ggplot(CGH_summary_daily, aes(mean_rt_lmp, color = factor(year))) +
  stat_ecdf() +
  labs(color = '',  x = 'Average RT LMP ($/MWh)',
       y = 'Empirical Cumulative Distribution Function') +
  theme_classic() +
  theme(legend.position = 'top')
```

Figure 3. Empirical Cumulative Distribution Function of RT LMP by Year (Chicago Gen Hub)[13]



[13] Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.

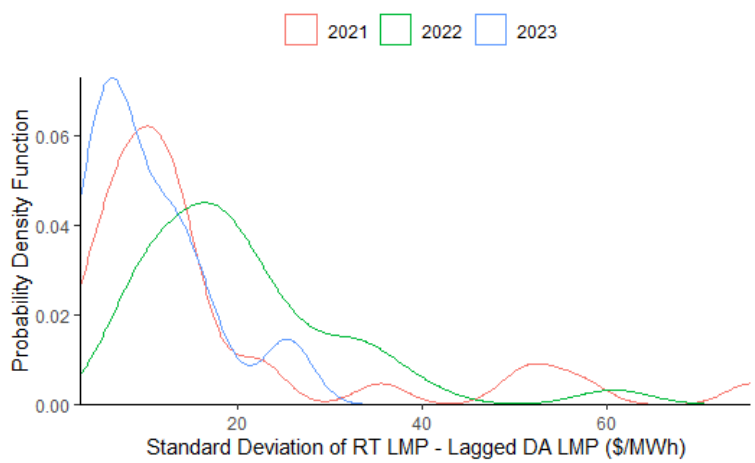
V. Pricing Anomalies and Patterns

18. The "spread" field in the PJM data reflects "total_lmp_rt" less "total_lmp_da." The code below calculates the volatility of this spread for each observation of the data, for CHICAGO GEN HUB, and plots the density of this volatility for August of each year. *In which year was the volatility of the LMP spread more variable? Use the PJM data to show how the distribution of volatility of the RT-DA LMP spread for the Chicago Gen Hub compares to that for the Eastern Hub ("EASTERN HUB").*

```
vol_daily_CGH <- data_CGH[, .(vol_spread = sd(spread, na.rm = TRUE)),
                      by = .(day, month, year)]

ggplot(vol_daily_CGH[month == 'Aug'], aes(x=vol_spread, color = factor(year))) +
  geom_density() +
  labs(x = 'Standard Deviation of RT LMP - DA LMP ($/MWh)', y = 'Probability Density
Function', color = '') +
  scale_x_continuous(expand = c(0,0)) +
  scale_y_continuous(expand = c(0,0)) +
  theme_classic() +
  theme(legend.position = 'top')
```

Figure 4. Distribution of August RT-DA LMP Spread Volatility by Year (Chicago Gen Hub)[14]



19. *Plot hourly RT LMP by day for the month of April 2023 (for Chicago Gen Hub).*[15] *Notice that RT LMP was often negative. What would negative prices mean? Try identifying other months or occasions for which RT LMP was negative and form a conjecture for why these negative prices occurred.*

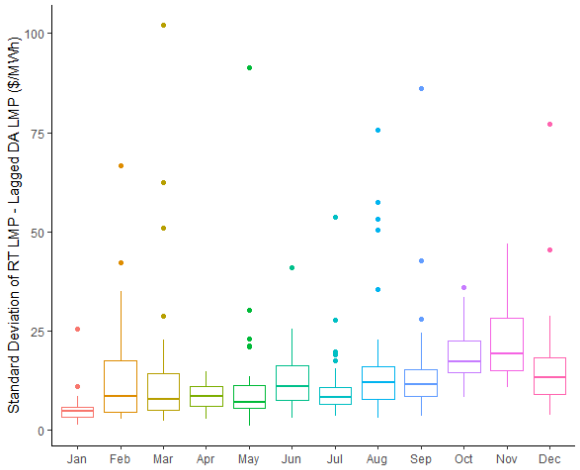
20. *Which hour of the day do you think RT LMP is highest and lowest? How would the answer depend on the season? Test your conjectures.*

[14] Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.
[15] Hint: consider using `geom_boxplot()`.

21. The code below plots boxplots of daily RT-DA LMP spread volatility for each month of 2021 for Chicago Gen Hub, such that individual points are outliers relative to the month, using the vol_daily_CGH dataset constructed in question 19. *Plot the same figure for 2022 and 2023. What day in 2021-2023 had the most extreme RT-DA LMP spread volatility?*

```
ggplot(vol_daily_CGH[year == 2021], aes(factor(month, levels = month.abb),
                                        vol_spread,
                                        color = factor(month, levels = month.abb))) +
  geom_boxplot() +
  labs(x = '', y = 'Standard Deviation of RT LMP - Lagged DA LMP ($/MWh)') +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust= 1)) +
  guides(color ='none') +
  theme_classic() +
  theme(legend.position = 'top')
```

Figure 5. Daily RT-DA LMP Spread Volatility by Month (2021, Chicago Gen Hub)[16]



[16] Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.

VI. How Weather Interacts with Energy Pricing

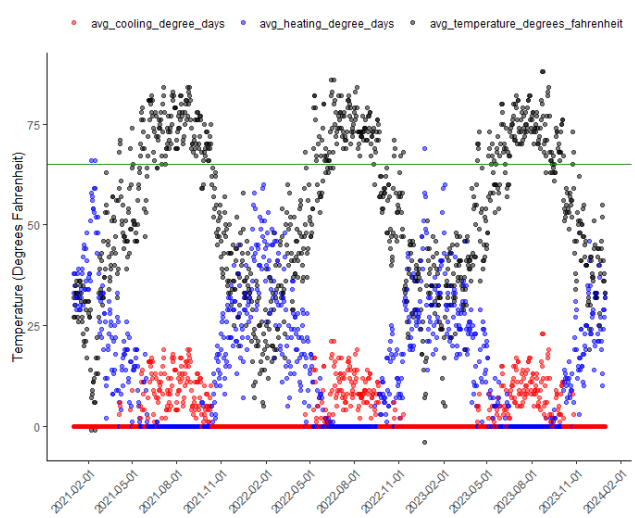
22. Let's turn to the dataset named "data_Chicago," which combines hourly LMP data from PJM (restricted to "CHICAGO GEN HUB") with daily weather data recorded at the Chicago O'Hare weather station. This dataset includes a number of weather-related fields, including average temperature, cooling degree days, and heating degree days. Heating degree days and cooling degree days respectively describe the number of degrees necessary to raise and lower the average temperature on a particular day to reach a fixed temperature (here, 65 degrees Fahrenheit). Note that if heating degree days is positive, then cooling degree days is zero—and vice versa. The code below plots heating degree days, cooling degree days, and average temperature for each day in 2021-2023 for Chicago. *What do you notice about the pattern of heating and cooling degree days throughout a year?*

```
daily_temp_Chicago <- data_Chicago[,.(date,month, year,
avg_temperature_degrees_fahrenheit, avg_cooling_degree_days, avg_heating_degree_days)]
%>%
  unique() %>%
  pivot_longer(cols = starts_with('avg'),
               names_to = 'temp_series',
               values_to = 'degrees')

setDT(daily_temp_Chicago)

ggplot(daily_temp_Chicago, aes(date, degrees, color = factor(temp_series))) +
  geom_point(alpha = 0.5) +
  scale_color_manual(values = c('red','blue','black')) +
  geom hline(yintercept = 65, color = 'forestgreen') +
  labs(x ='', color = '', y = 'Temperature (Degrees Fahrenheit)') +
  scale_x_date(date_breaks = '3 months') +
  theme_classic() +
  theme(legend.position = 'top',
        axis.text.x = element_text(angle = 45, vjust = 1, hjust= 1))
```

Figure 6. Chicago Daily Average Temperature (2021 – 2023)[17]



[17] Weather data from Hitachi Energy Velocity Suite.

23. The code below plots the probability density function (PDF) of average daily temperature by year, using the "daily_temp_Chicago" dataset constructed above. *What do you notice about the shape of temperature distributions? Form a conjecture explaining the shape and test it by comparing the PDFs by month using by adding the following command at the end of the code:* `facet_wrap(~factor(month, levels = month.abb))`.

```
ggplot(daily_temp_Chicago[temp_series == 'avg_temperature_degrees_fahrenheit'],
       aes(degrees, color = factor(year))) +
  geom_density() +
  labs(color ='', x = 'Average Temperature (Degrees Fahrenheit)',
       y = 'Probability Density Function') +
  theme_classic() +
  theme(legend.position = 'top')
```

24. The following code produces yearly ECDFs of average daily temperature and average daily RT LMP for Chicago. *What do you notice about the relationship, in terms of FOSD, between ECDFs for the LMPs? Does the same relationship hold for temperature? Conjecture as to why temperature patterns may or may not have translated to LMP patterns. Try the same exercise, comparing by month.*

```
lmp_temp_Chicago <- data_Chicago[,.(mean_lmp_rt = mean(total_lmp_rt),
                                    mean_lmp_da = mean(total_lmp_da),
                                    vol_spread = sd(spread),
                                    temp = mean(avg_temperature_degrees_fahrenheit),
                                    humidity = mean(relative_humidity_percent)),
                                 by = .(date, month, year)]

ggplot(lmp_temp_Chicago, aes(mean_lmp_rt, color = factor(year))) +
  stat_ecdf() +
  labs(x = 'Daily Average RT LMP ($/MWh)', y = 'Empirical Cumulative Distribution
Function',
       color = '') +
  theme_classic() +
  theme(legend.position = 'top')

ggplot(lmp_temp_Chicago, aes(temp, color = factor(year))) +
  stat_ecdf() +
  labs(x = 'Daily Average Temperature (Degrees Fahrenheit)', y = 'Empirical Cumulative
Distribution Function',
       color = '') +
  theme_classic() +
  theme(legend.position = 'top')
```

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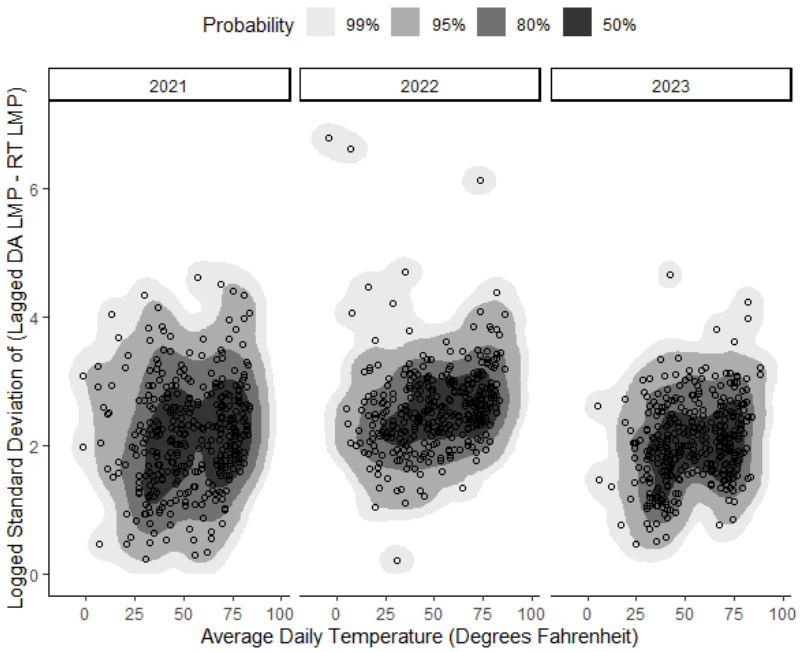
25. The code below plots the joint density of average daily temperature and the logged standard deviation of the daily volatility of the RT-DA LMP spread. *Is there an obvious pattern that emerges? Try identifying notable outliers in the lmp_temp_Chicago dataset. Conjecture as to why outliers emerged on those days.*

```
ggplot(lmp_temp_Chicago, aes(x=temp, y=log(vol_spread)) ) +
  geom_hdr(xlim = c(-25, 100), ylim = c(0,10)) +
  geom_point(shape = 21) +
  labs(fill = '', alpha = 'Probability',
       x = 'Average Daily Temperature (Degrees Fahrenheit)',
       y = 'Logged Standard Deviation of (Lagged DA LMP - RT LMP)') +
  theme_classic() +
  theme(legend.position = 'top') +
  facet_grid(~year)
```

Figure 7. Average Daily Temperature vs. Volatility of Daily DA-RT LMP Spread (Chicago)



26. The code below renders the plots above using a 3D density plot. *Give it a try*! What *do the peaks and valleys represent*?

```
dens_Chicago <- kde2d(lmp_temp_Chicago[date>'2021-01-
01']$temp,log(lmp_temp_Chicago[date>'2021-01-01']$vol_spread))

plot_ly(x = dens_Chicago$x, y=dens_Chicago$y, z = dens_Chicago$z) %>%
  add_surface() %>%
  layout(scene = list(xaxis = list(title = 'Avg. Temperature'),
                      yaxis = list(title = 'Log.SD (DA LMP - RT LMP)'),
                      zaxis = list(title = 'PDF')))
```

VII. Comparing Interconnected Transmission Regions

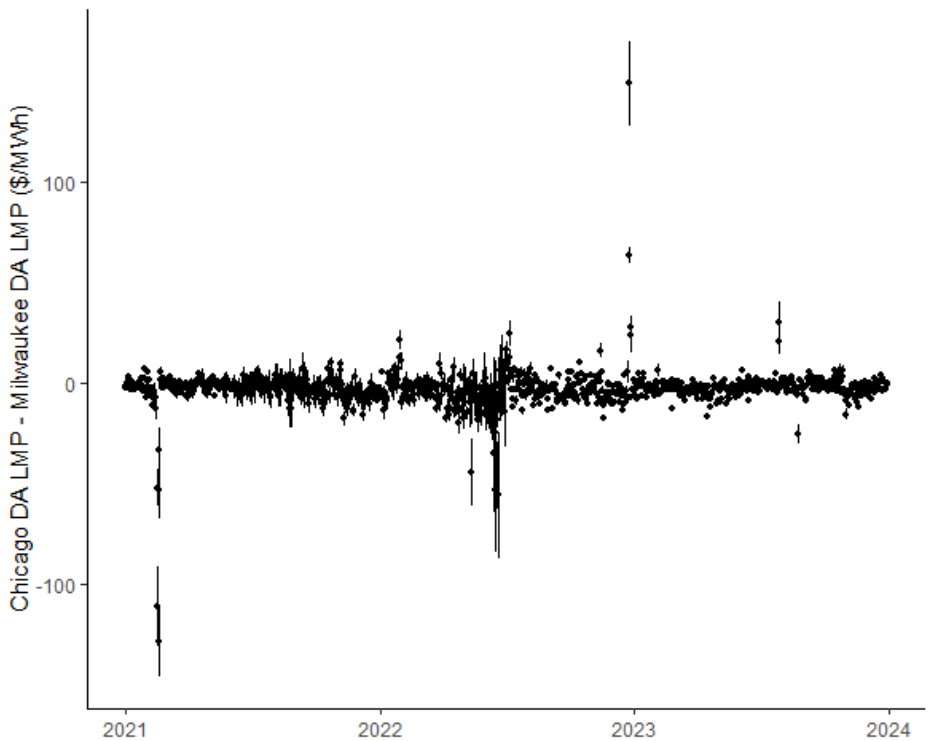
The states west of PJM belong to the Midcontinent Independent System Operator (MISO). Milwaukee and Chicago, two cities merely 90 minutes apart by car, belong to two different ISOs (Milwaukee in MISO and Chicago in PJM). In this section, we explore differences in LMPs and their responsiveness to weather patterns, in two separate (yet interconnected) parts of the grid.

27. The code below plots the spread between average daily DA LMPs in Chicago and Oak Creek Power Plant (near Milwaukee). *Consider the outlier days. Could they be explained by variations in weather? Plot the same figure for the spread in average daily temperatures to test your hypothesis.*

```
lmp_cities <- data.table(
  date = data_Milwaukee$date,
  Milwaukee = data_Milwaukee$DA_LMP,
  Chicago = data_Chicago$total_lmp_da,
  da_spread =  data_Chicago$total_lmp_da - data_Milwaukee$DA_LMP
)

ggplot(lmp_cities, aes(x=date,y=da_spread)) +
  stat_summary(size=0.2) +
  labs(x='',y='Chicago DA LMP - Milwaukee DA LMP ($/MWh)')+
  theme_classic()
```

Figure 8. Spread Between Average Daily DA LMPs in Chicago vs Milwaukee[18]



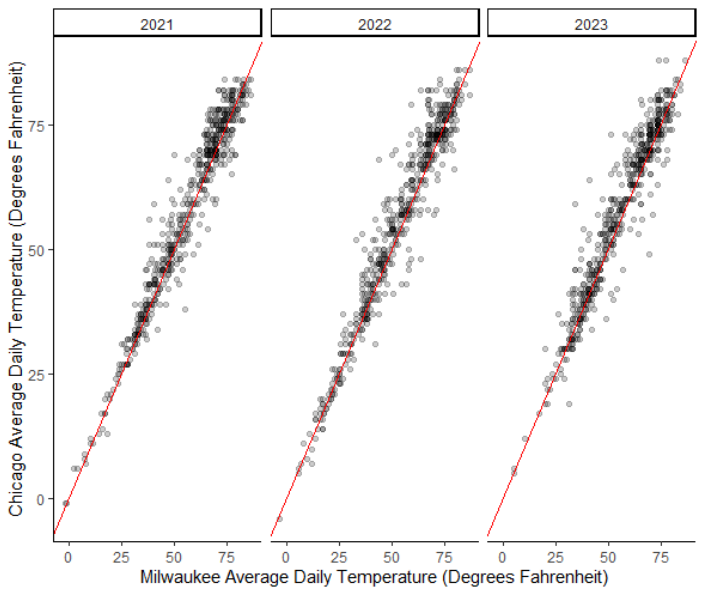
[18] Chicago LMP data from Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition. Milwaukee DA LMP data from MISO Market Reports: Historical Annual Day-Ahead LMPs.

28. One might expect that weather in Milwaukee and Chicago to be similar. The code below generates scatter plots comparing average daily temperatures in Milwaukee and Chicago, showing temperatures in the two cities are closely aligned. *Try creating a similar plot for average daily DA LMPs in the two cities. How closely aligned are the average daily DA LMPs in the two cities? Why might LMPs in the two cities differ more than temperatures?*

```
temperature_cities <- data.table(
  date = data_Milwaukee$date,
  Milwaukee = data_Milwaukee$avg_temp_f,
  Chicago = data_Chicago$avg_temperature_degrees_fahrenheit) %>%
unique()

ggplot(temperature_cities, aes(x = Milwaukee, y = Chicago)) +
  geom_point(alpha = 0.2) +
  geom_abline(intercept = 0, slope = 1,color='red') +
  labs(x='Milwaukee Average Daily Temperature (Degrees Fahrenheit)',
       y='Chicago Average Daily Temperature (Degrees Fahrenheit)',
       color = '') +
  facet_wrap(~year(date)) +
  theme_classic()
```

Figure 9. Average Daily Temperatures (Milwaukee vs. Chicago)[19]

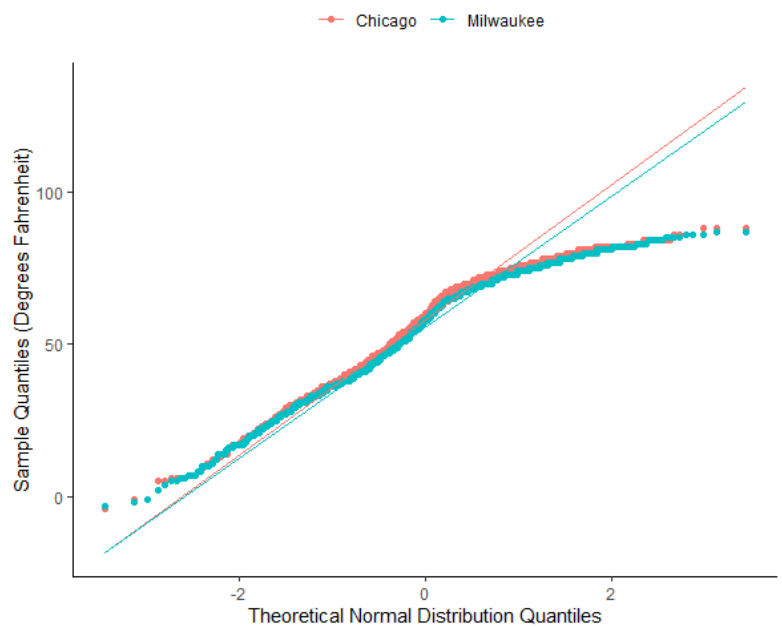


[19] Chicago and Milwaukee weather data from Hitachi Energy Velocity Suite.

29. We can examine whether daily temperatures in Chicago and Milwaukee are drawn from a normal distribution using a Q-Q plot. *Based on the Q-Q plot created from the code below, to what extent are these temperature series normal? Try creating a similar Q-Q plot comparing Austin and Corpus Christi temperatures to the normal distribution. As an optional step, try creating a Q-Q plot of Chicago average temperatures against Milwaukee temperatures. Are these series drawn from the same distribution?*

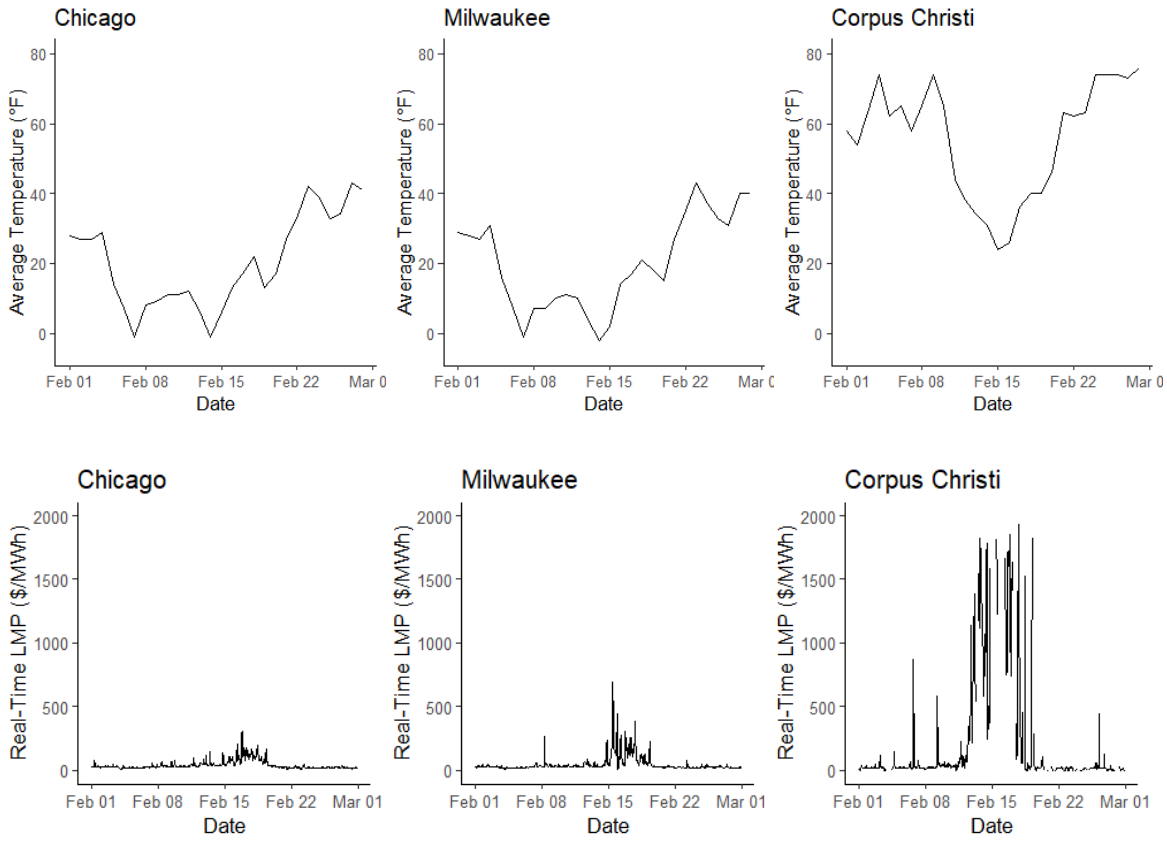
```
ggplot(temperature_cities %>% pivot_longer(!date, names_to = 'series', values_to =
'temp'),
       aes(sample = temp, color = series)) +
  geom_qq() +
  geom_qq_line() +
  labs(x='Theoretical Normal Distribution Quantiles',
       y='Sample Quantiles (Degrees Fahrenheit)',
       color = '') +
  theme_classic() +
  theme(legend.position = 'top')
```

Figure 10. Q-Q Plots of Chicago and Milwaukee Daily Average Temperature Against Standard Normal Distribution



30. In February 2021, Texas suffered a devastating winter storm that exacerbated the vulnerabilities of its power grid. The code below displays average daily temperatures and real-time LMPs in Chicago, Milwaukee, and Corpus Christi during this month. *Note the scale for each plot. What do you observe from these plots? Conjecture as to the relationship between prices and weather. Do you think the relationship is linear? Describe the role and nuances of anomalous weather.*

Figure 11. Average Daily Temperatures and Hourly RT LMPs in 2021 (Chicago, Milwaukee, and Corpus Christi)[20,21]



[20] Chicago LMP data from Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition. Milwaukee RT LMP data from MISO Market Reports: Historical Annual Real-Time LMPs. Corpus Christi LMP data and all weather data from Hitachi Energy Velocity Suite.

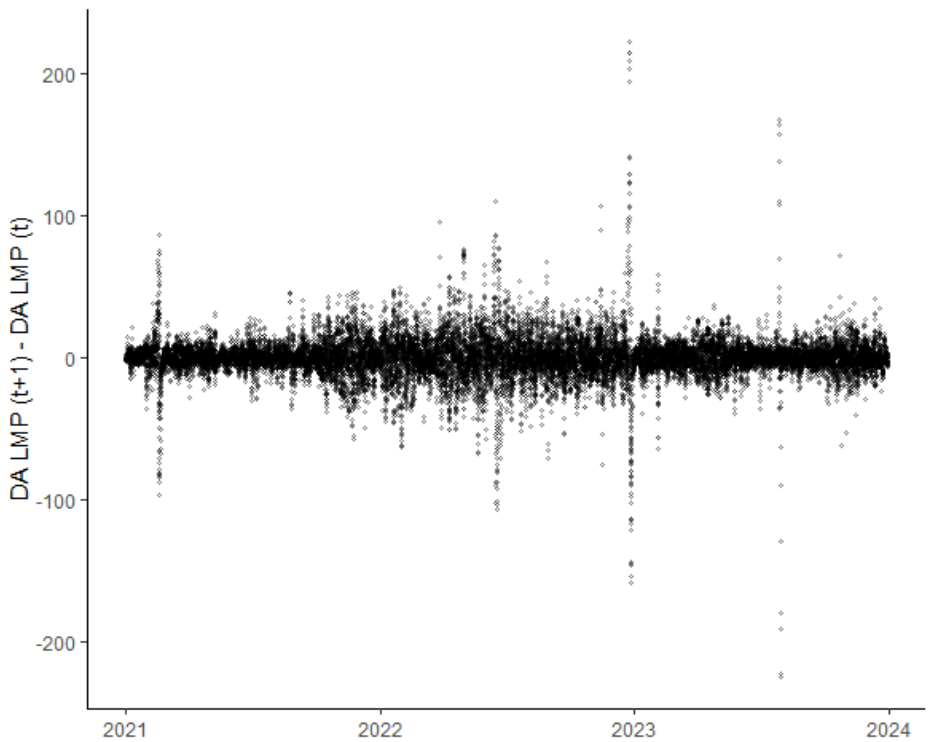
[21] Although the maximum value of the pricing graphs' y-axis is $2,000/MWh, RT LMPs in Corpus Christi exceeded $9,000/MWh in February 2021.

VIII. Contracts for Differences

31. The code below plots day-over-day changes in the Chicago hourly DA LMP. *What does each point on the plot represent? Why are the observations on for most days approximately symmetric around zero? Which days exhibited the largest day-over-day changes in DA LMP? How do day-to-day changes in DA LMP relate to contracts for differences (CfDs)?*

```
ggplot(data_Chicago[order(date)] %>%
         .[,prev_lmp_da := shift(total_lmp_da, type = 'lag'), by = hour],
       aes(x = date, y = total_lmp_da - prev_lmp_da)) +
  geom_point(shape = 21, size = 0.8, alpha = 0.5) +
  labs(x = '',y = 'DA LMP (t+1) - DA LMP (t)') +
  theme_classic()
```

Figure 12. Day-over-Day Changes in Chicago DA LMP ($/MWh)[22]



32. *Form your own hypotheses about how day-over-day changes in DA LMP may vary according to different dimension of the Chicago data (e.g., day-of-week, temperature changes, etc.). Test them. Discuss how patterns in day-over-day changes in DA LMP may relate to energy storage needs.*

[22] Chicago LMP data from Data Miner 2, PJM, at https://dataminer2.pjm.com/feed/rt_da_monthly_lmps/definition.

33. The code below creates a data set containing the ratio of day-over-day DA LMP spreads between Chicago and Milwaukee. That is, it computes the day-over-day change in hourly DA LMP for Chicago and Milwaukee and then computes their ratio in the `spread_ratio` field. Use this data set to plot `spread_ratio` and interpret the magnitude of `spread_ratio`. What do larger values mean and how do they relate to incentives for energy transmission and storage given your interpretation, what has changed in 2023 relative to 2021 and 2022?

```
internodal_spread <- data.table(date = data_Chicago$date,
                                hour = data_Chicago$hour,
                                day_of_week = data_Chicago$day_of_week,
                                month = data_Chicago$month,
                                da_Chicago = data_Chicago$total_lmp_da,
                                da_Milwaukee = data_Milwaukee$DA_LMP,
                                da_CorpusChristi = data_CorpusChristi$DA_LMP) %>%
  .[order(date)] %>%
  .[,':=' (prev_da_Chicago =  shift(da_Chicago, fill = NA, type = 'lag'),
           prev_da_Milwaukee = shift(da_Milwaukee, fill = NA, type = 'lag'),
           prev_da_CorpusChristi = shift(da_CorpusChristi, fill = NA, type = 'lag')),
    by = hour] %>%
  .[, ':=' (Chicago_da_spread = da_Chicago - prev_da_Chicago,
            Milwaukee_da_spread = da_Milwaukee - prev_da_Milwaukee,
            CorpusChristi_da_spread = da_CorpusChristi - prev_da_CorpusChristi)] %>%
  .[, spread_ratio := Chicago_da_spread/Milwaukee_da_spread]
```

34. In the previous question, we explored the ratio of day-over-day DA LMP spreads between Milwaukee and Chicago. The code below creates a similar dataset for the ratio of day-over-day DA LMP spreads between Chicago and Corpus Christi, TX; the `spread_ratio_ercot` variable holds this data (Corpus Christi belongs to the Electric Reliability Council of Texas, or "ERCOT"). What do larger values mean, and how do they relate to incentives for energy transmission and storage?

```
internodal_spread$spread_ratio_ercot <- internodal_spread$Chicago_da_spread /
internodal_spread$CorpusChristi_da_spread
```

35. The code below tells you the number of days that `spread_ratio` and `spread_ratio_ercot` were above 1, respectively, between 2021 and 2023. *Interpret these results.*

```
count1 <- sum(internodal_spread$spread_ratio > 1, na.rm = TRUE)
count2 <- sum(internodal_spread$spread_ratio_ercot > 1, na.rm = TRUE)
print(count1)
print(count2)
count3 <- sum(internodal_spread$spread_ratio > 0, na.rm = TRUE)
count4 <- sum(internodal_spread$spread_ratio < 0, na.rm = TRUE)
print(count3)
print(count4)
```

36. Construct a spread metric that considers the contemporaneous spread between Chicago and Milwaukee to those from the preceding day. Interpret this ratio. How does it differ from `spread_ratio`? Try comparing these two different series using a plot.

IX. Load and Price Duration Curves

37. Use the code below to load a dataset, created by Bolarinwa Ajanaku,[23] containing daily observations of PJM data from January 1, 2016 to December 31, 2019.[24] This dataset includes a number of series, including load, RT LMP, as well as supply and prices for natural gas and wind.

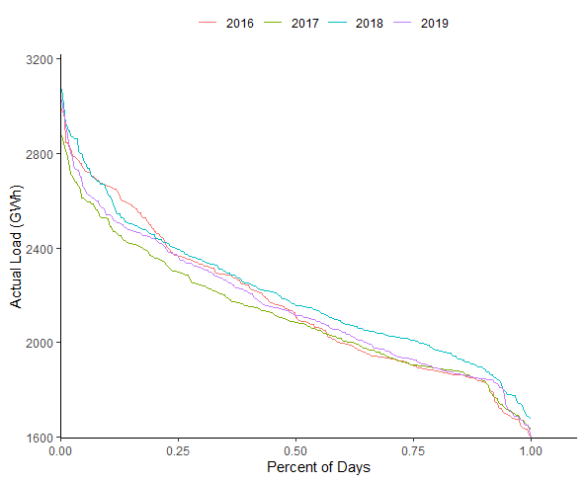
```
daily_PJM <- fread('PJM Daily.csv')
```

38. The code below generates annual load duration curves for PJM, over the years 2016-2019. *Based on the plot, which years featured higher demand for electricity? Try adapting the code to annual price duration curves for PJM, using the* `rtm_price` *variable in place of* `Actual_load_GWh`*. Which year featured the highest LMP prices? Is this consistent with your inference based on the load duration curve?*

```
ldc <- daily_PJM[, .(Date, Actual_load_GWh)] %>%
  .[, Date := as.Date(Date, format = "%m/%d/%Y")] %>%
  .[, rank := frank(-Actual_load_GWh) / .N, by = year(Date)]

ggplot(ldc, aes(rank, Actual_load_GWh, color = factor(year(Date)))) +
  geom_line() +
  scale_x_continuous(expand = expand_scale(mult=c(0,0.1)), limits = c(0,NA))+
  scale_y_continuous(expand = expand_scale(mult=c(0,0.1))) +
    labs(x = 'Percent of Days', y = 'Actual Load (GWh)', color = '',
       title = 'PJM Annual Load Duration Curves') +
  theme_classic() +
  theme(plot.title = element_text(hjust=0.5),
        legend.position = 'top',
        plot.caption = element_text(hjust = 0))
```

Figure 13. Annual PJM Load Duration Curves[25]



[23] https://baajanaku.wixsite.com/bolarinwa.
[24] For the online source of the data, see here: https://data.mendeley.com/datasets/b4rwv95dzb/1.
[25] Daily PJM data from https://data.mendeley.com/datasets/b4rwv95dzb/1.

39. *Try plotting load duration curves for Austin, Milwaukee, and Corpus Christi, using the "load_mw" field. Which cities typically have higher loads?*

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